

Topic Modelling & Latent Dirichlet Allocation (LDA)

STQD 6114: Unstructured Data Analytics

What is topic modelling?

- a statistical model to discover the topics that occur in a collection of documents
- method for finding a group of words (ie topics) from a collection of document that best represents the info in the collection
- it is a form of text mining

Why?

- Data exploration
- Rough idea on what is the structure/pattern/category of your text data
- Allow us to answer big picture questions quickly, & without human interventions

Example

% of Gazette	Most likely words in a topic in order of likelihood	Human-added topic label
5.6	away reward servant named feet jacket high paid hair coat run inches master...	<i>Runaways</i>
5.1	state government constitution law united power citizen people public congress...	<i>Government</i>
4.6	good house acre sold land meadow mile premise plantation stone mill dwelling...	<i>Real Estate</i>
3.9	silk cotton ditto white black linen cloth women blue worsted men fine thread...	<i>Cloth</i>

Example

- Topic modeling as
 - Qualitative Social Evidence
 - angry speech and patriotism over time
 - similar trend
 - Literary Theoretical Springboard
 - study on poem
 - the opaque features in a poem harden the machine's task; in which human assessment is still needed
 - novel/book genre

More examples

- Suppose you have these statements:

- d1* • I love father
- d2* • Mommy plays with sister
- d3* • My hamster is cute
- d4* • Sister likes rabbit

 topic A
 topic B

- Latent Dirichlet Allocation: a method to automatically discovering topics for a set of statements
- Referring to the example, and if you asked for 2 topics, LDA might produce the following results:

- Sentences 1 and 2: 100% Topic A
- Sentence 3: 100% Topic B
- Sentence 4: 50% Topic A & 50% Topic B

*need to identify which
topic based on word
collected*

The LDA model - How?

- LDA represents documents as mixtures of topics that contains words with certain probabilities
- Assume that the document is produced based on this process:
 - The number of words, N is decided; and distributed as Poisson (example)
 - Choose a topic mixture (according to Dirichlet distribution over a fixed set of K topics): for example; $2/3$ on family and $1/3$ on pet

- for each word, use the topic to generate the word; based on multinomial distribution. For example; with topic family, we might generate “mommy” with 30% probability, daddy 40% and so on
- then, LDA will try to backtrack this process to find a set of topics that are likely to have generated the collection
- hence, topic modeling is a way of extrapolating backward from a collection of documents to infer the topics that could have generated them

- Further reading:

<http://www.jmlr.org/papers/volume3/blei03a/blei03a.pdf>

Example

- Say that we pick 4 to be the number of words in our document
- Then, we decide that D will be $\frac{1}{2}$ about family and $\frac{1}{2}$ about pet
- Pick the first word from the family topic; $\rightarrow$ mommy
- Pick the second word from the pet topic; $\rightarrow$ hamster
- Pick the third word from the family topic; $\rightarrow$ sister
- Pick the fourth word from the pet topic; $\rightarrow$ rabbit
- Hence, the document generated will consists of these words:
mommy-hamster-sister-rabbit

Expected output from LDA

- List of terms in each topic

	Topic 1	Topic 2	Topic 3	Topic 4	Topic 5
1	work	question	chang	system	project
2	practic	map	organ	data	manag
3	mani	time	consult	model	approach

List of the document to the (primary) topic

	Document	Topic
d_1	BeyondEntitiesAndRelationships.txt	4
d_2	bigdata.txt	4
d_3	ConditionsOverCauses.txt	5
d_4	EmergentDesignInEnterpriseIT.txt	4
d_5	FromInformationToKnowledge.txt	2
d_6	FromTheCoalface.txt	1

$K = 5$

- Topic probabilities by documents

		Topic 1	Topic 2	Topic 3	Topic 4	Topic 5
d_1	BeyondEn	0.071	0.064	0.024	0.741	0.1
d_2	bigdata.	0.182	0.221	0.182	0.26	0.156
d_3	Conditio	0.144	0.109	0.048	0.205	0.494
d_4	Emergent	0.121	0.226	0.204	0.236	0.213
d_5	FromInfo	0.096	0.643	0.026	0.169	0.066

LDA in R

- Data source & R code:

<https://cran.r-project.org/web/packages/topicmodels/vignettes/topicmodels.pdf>

<https://eight2late.wordpress.com/2015/09/29/a-gentle-introduction-to-topic-modeling-using-r/>

<https://www.tidyttextmining.com/topicmodeling.html>

Example in R – topicmodels package

pre processing .

- prepping the data: transform to lower case, remove symbols, punctuations, general errors (different versions of English, stopwords)
- create a document term matrix *dtm*
- frequency of each word *freq*

Example in R

```
library(topicmodels)

data("AssociatedPress")

AssociatedPress
```

```
## <<DocumentTermMatrix (documents: 2246, terms: 10473)>>
## Non-/sparse entries: 302031/23220327
## Sparsity           : 99%
## Maximal term length: 18
## Weighting           : term frequency (tf)
```

```
# set a seed so that the output of the model is predictable  
ap_lda <- LDA(AssociatedPress, k = 2, control = list(seed = 1234))  
ap_lda
```

```
## A LDA_VEM topic model with 2 topics.
```



```
library(tidytext)

ap_topics <- tidy(ap_lda, matrix = "beta")
ap_topics
```

```
## # A tibble: 20,946 x 3
##   topic term      beta
##   <int> <chr>    <dbl>
## 1      1 aaron  1.69e-12
## 2      2 aaron  3.90e- 5
## 3      1 abandon 2.65e- 5
## 4      2 abandon 3.99e- 5
## 5      1 abandoned 1.39e- 4
## 6      2 abandoned 5.88e- 5
## 7      1 abandoning 2.45e-33
```

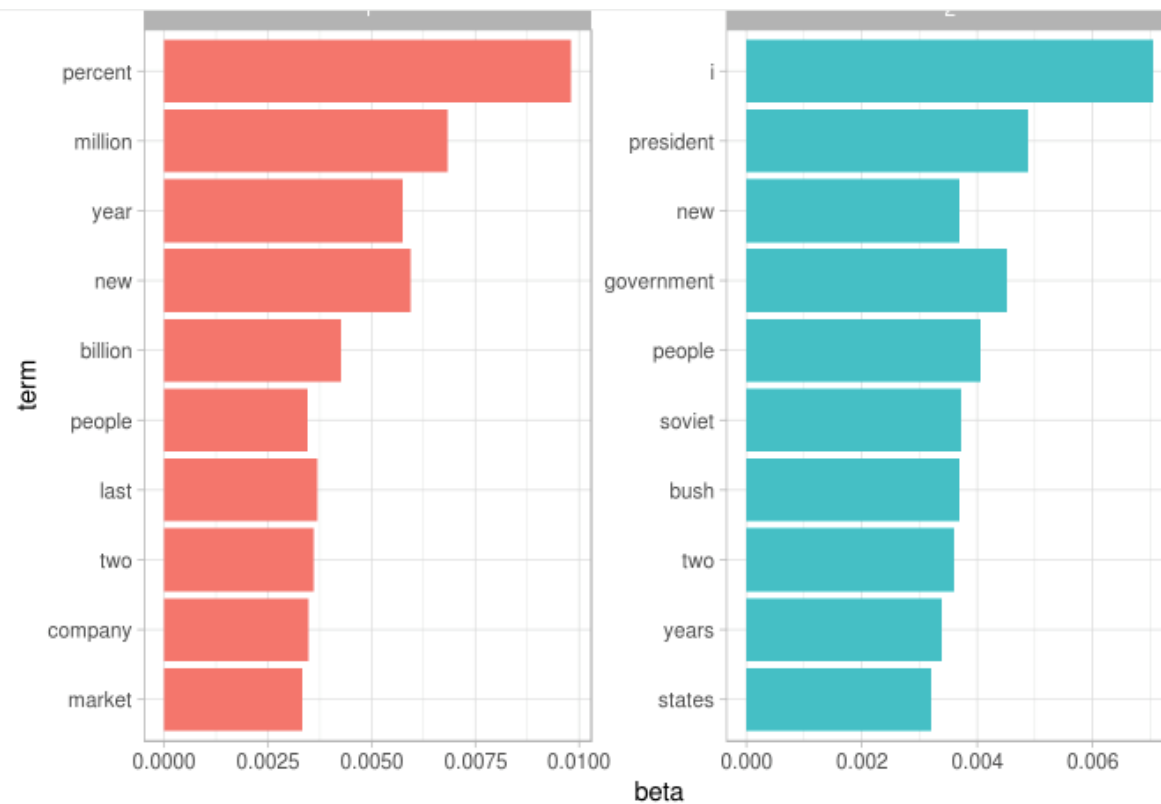
extracting the
per-topic-per-
word
probabilities

```
library(ggplot2)
library(dplyr)

ap_top_terms <- ap_topics %>%
  group_by(topic) %>%
  top_n(10, beta) %>%
  ungroup() %>%
  arrange(topic, -beta)

ap_top_terms %>%
  mutate(term = reorder(term, beta)) %>%
  ggplot(aes(term, beta, fill = factor(topic))) +
  geom_col(show.legend = FALSE) +
  facet_wrap(~ topic, scales = "free") +
  coord_flip()
```

find the 10
terms that are
most common
within each
topic



This visualization lets us understand the two topics that were extracted from the articles. The most common words in topic 1 include “percent”, “million”, “billion”, and “company”, which suggests it may represent business or financial news. Those most common in topic 2 include “president”, “government”, and “soviet”, suggesting that this topic represents political news. One important observation about the words in each topic is that some words, such as “new” and “people”, are common within both topics. This is an advantage of topic modeling as opposed to “hard clustering” methods: topics used in natural language could have some overlap in terms of words

the terms that had the greatest difference in β between topic 1 and topic 2. This can be estimated based on the log ratio of the two:

$\log(\beta_2/\beta_1)$. A log ratio is useful because it makes the difference symmetrical:

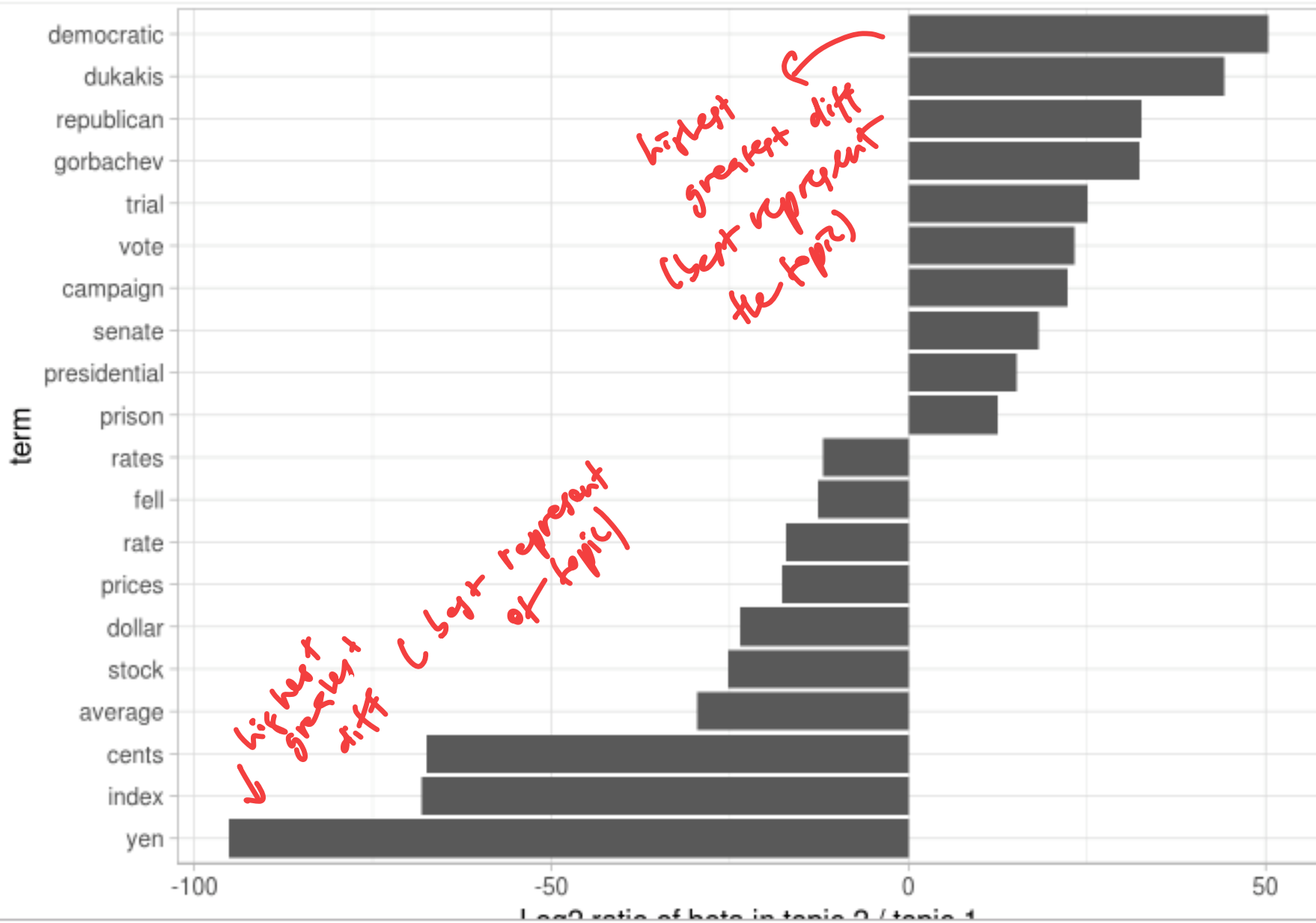
β_2 being twice as large leads to a log ratio of 1, while β_1 being twice as large results in -1). To constrain it to a set of especially relevant words, we can filter for relatively common words, such as those that have a β greater than 1/1000 in at least one topic.

```
library(tidyr)

beta_spread <- ap_topics %>%
  mutate(topic = paste0("topic", topic)) %>%
  spread(topic, beta) %>%
  filter(topic1 > .001 | topic2 > .001) %>%
  mutate(log_ratio = log2(topic2 / topic1))

beta_spread
```

filter
accordingly
when value
too small



```
ap_documents <- tidy(ap_lda, matrix = "gamma")  
ap_documents
```

↓
find the
probability

Check doc that's
belong to what
topic and its
probability.

```
## # A tibble: 4,492 x 3  
##   document topic   gamma probability  
##   <int> <int>   <dbl>  
## d1      1      1 0.248  
## d2      2      1 0.362
```

document-topic
probabilities

```
tidy(AssociatedPress) %>%  
  filter(document == 6) %>%  
  arrange(desc(count))
```

```
## # A tibble: 287 x 3  
##   document term      count  
##   <int> <chr>    <dbl>  
## 1      6 noriega      16  
## 2      6 panama      12  
## 3      6 jackson       6  
## 4      6 powell        6  
## 5      6 administration  5  
## 6      6 economic        5
```

list of words for a specific
document

Text Analysis I: LDA

```
library(tidytext)
```

```
library(topicmodels)
```

```
library(tidyr)
```

```
library(ggplot2)
```

```
library(dplyr)
```

```
data("AssociatedPress")
```

```
ap_lda<-LDA(AssociatedPress,k=2,control=list(seed=1234)) #create two-topic LDA model
```

```
ap_topics<-tidy(ap_lda,matrix="beta") #Extract the per-topic-per-word-probabilities
```

```
#Find terms that are most common within each topics
```

```
ap_top_terms <- ap_topics %>% group_by(topic) %>% top_n(10,beta) %>% ungroup () %>% arrange (topic, -beta)
```

```
ap_top_terms%>% mutate(term=reorder(term,beta))%>%  
ggplot(aes(term,beta,fill=factor(topic)))+geom_col(show.legend=FALSE)+
```

```
facet_wrap(~topic,scales="free")+coord_flip() #visualize the above
```

```
beta_spread <- ap_topics %>% mutate (topic=paste0("topic",topic)) %>% spread(topic,beta) %>%
```

```
filter (topic1>0.001 | topic2 > 0.001) %>% mutate(log_ratio = log2(topic2/topic1))
```

```
beta_spread%>% mutate(term=reorder(term,log_ratio))%>%  
ggplot(aes(term,log_ratio))+geom_col(show.legend=FALSE)+coord_flip()
```

```
ap_documents<-tidy(ap_lda,matrix="gamma") #Extract the per-document-per-topic-probabilities
```

```
ap_documents
```

```
tidy(AssociatedPress)%>%filter(document==6)%>%arrange(desc(count)) #Check the most common words in the document, eg  
document 6
```


Exercise

- Perform the LDA analysis to your own choice of data. Interpret the results.